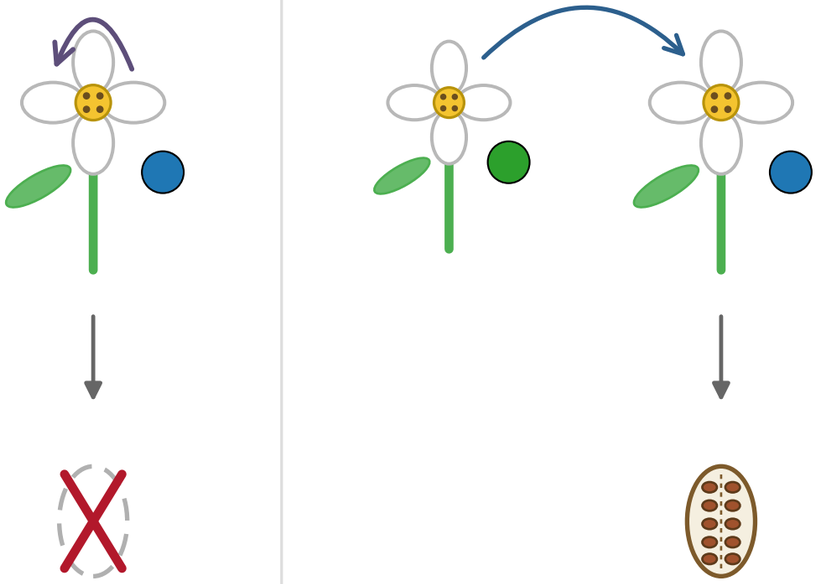
1. Self-incompatible: plants reject their own pollen

color = S-type
(32 known in LEPA)

self-incompatible

own pollen

other S-type



no seed

seed

the S-locus (a male + a female gene) gives each plant an S-type: pollen with the flower's own S-type is rejected

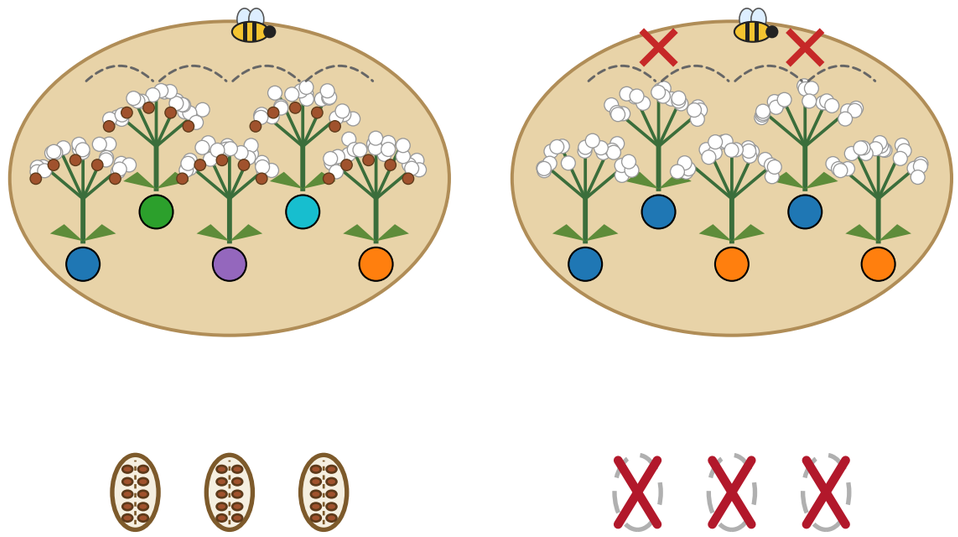
2. Fragmentation + drift → mate limitation → population decline

mate limitation

habitat fragmentation
+ genetic drift

many S-types

few S-types

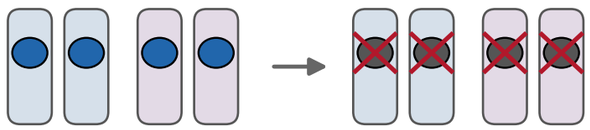


in the field: less seed than plant size predicts

small, isolated populations lose S-types by drift:
plants run out of compatible mates and set little seed

3. Rapid adaptation: escape from SI → self-compatible, short-term gain

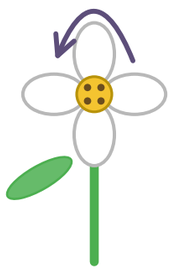
self-compatible



working S-locus
(4 copies)

broken copies

own pollen



LEPA — 263 plants genotyped

247 self-incompatible

15 partly broken

1 self-compatible

in the field: more seed than plant size predicts



seed

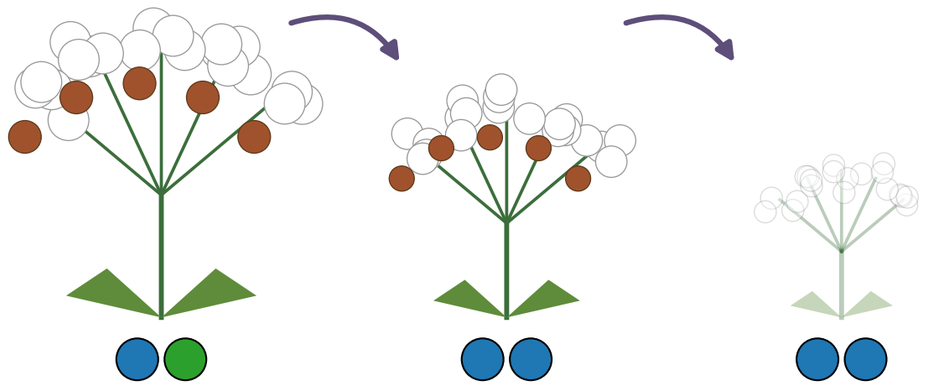
plants with broken S-locus copies accept their own pollen and set seed alone, without mates

4. Long-term cost: inbreeding depression → population decline

inbreeding

self

self



parent

selfed
offspring

next
generation

seed ↓

survival ↓

diversity ↓

selfed offspring grow more alike and weaker,
generation after generation